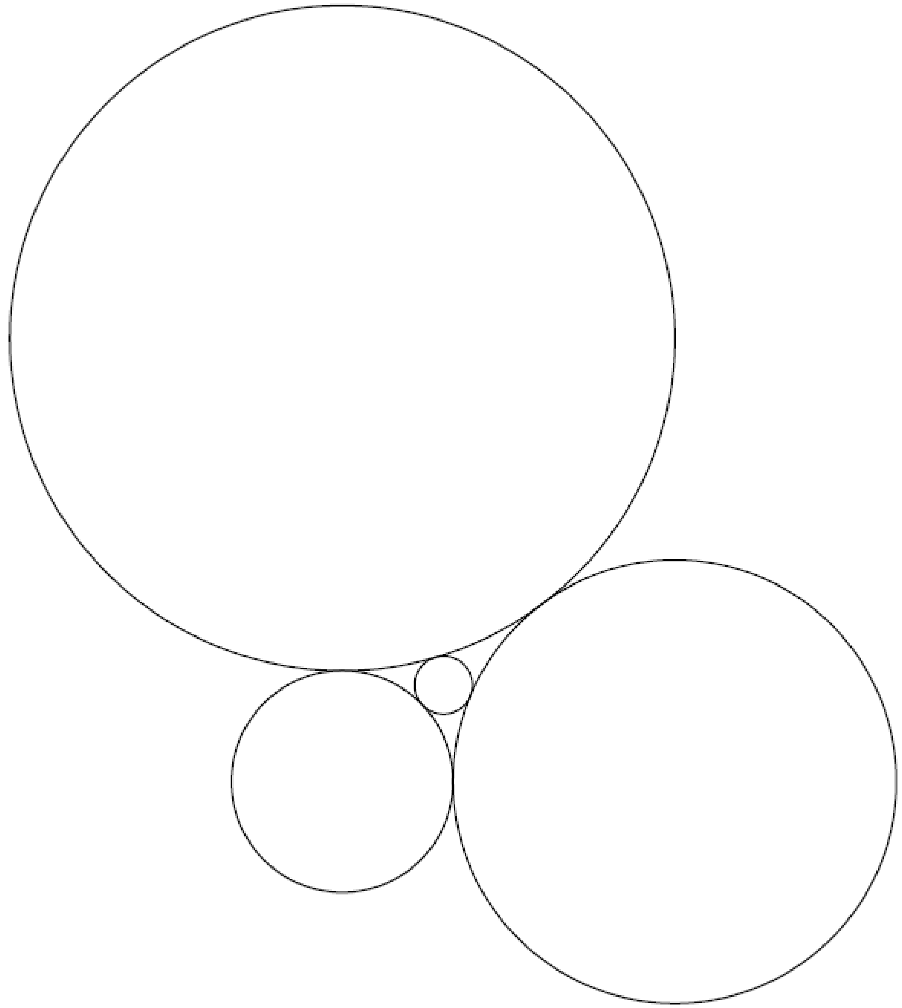
2025 AMC 10A Problem 22 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 22

Problem:

A circle of radius r is surrounded by three circles, whose radii are $1, 2,$ and $3,$ all externally tangent to the inner circle and to each other, as shown.



What is r?

Answer Choices:

A. $\frac{1}{4}$

B. $\frac{6}{23}$

C. $\frac{3}{11}$

D. $\frac{5}{17}$

E. $\frac{3}{10}$

Solution:
Let the centers of the circles with radii $r, 1, 2,$ and 3 be $O, A, B,$ and C respectively.
Since

$$AB = 1 + 2 = 3, \quad AC = 1 + 3 = 4, \quad BC = 2 + 3 = 5,$$

the triangle ABC is a right-angled triangle with the right angle at B. Place coordinates so that

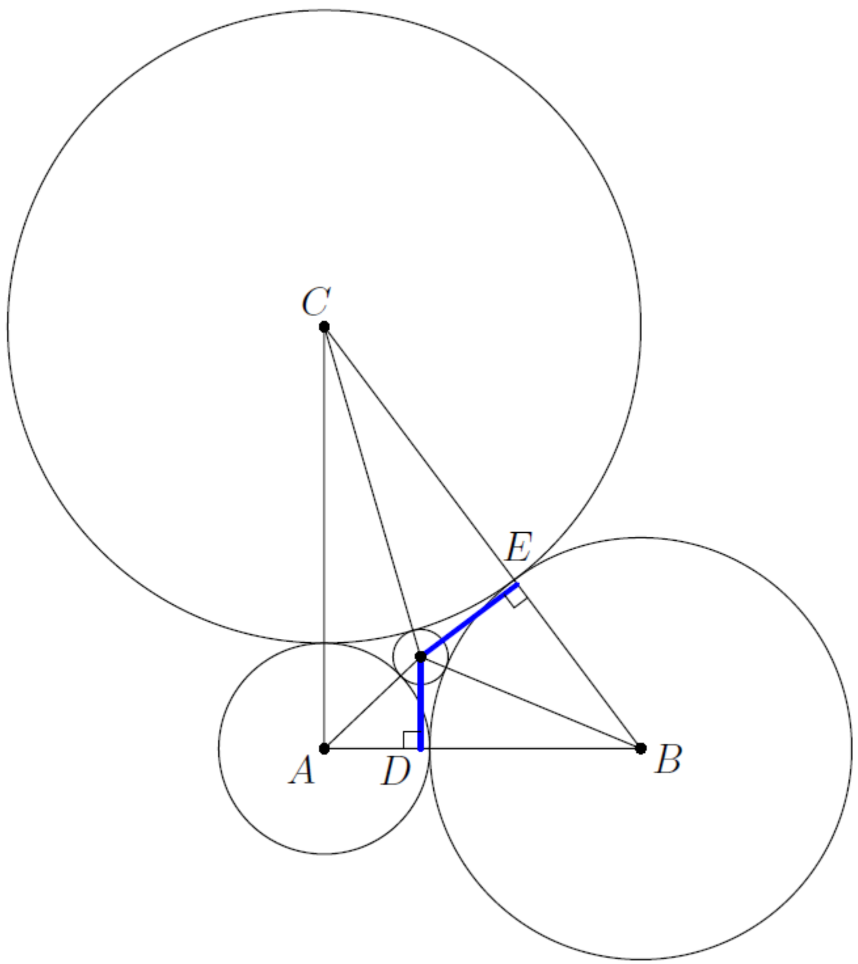
$$A(0,0), \quad B(3,0), \quad C(3,4).$$

Let D be the foot of the perpendicular from O to $\overline{AB}$, and E be the foot from O to $\overline{BC}$.
Set

$$OD = y, \quad OE = x,$$

so that O has coordinates $(3 - x, y)$. Since each circle is tangent externally to its neighbors,

$$OA = r + 1, \quad OB = r + 2, \quad OC = r + 3.$$



Therefore, using the $\text{Pythagorean Theorem}$:

$$\begin{aligned} OA^2 &= x^2 + y^2 = (r+1)^2 && (1) \\ OB^2 &= (3-x)^2 + y^2 = (r+2)^2 && (2) \\ OC^2 &= x^2 + (y-4)^2 = (r+3)^2 && (3) \end{aligned}$$

Subtract (2) from (1) and (3) to eliminate y^2 and x^2 respectively:

$$(1)-(2):\ (3-x)^2-x^2=(r+2)^2-(r+1)^2 \implies 9-6x=2r+3 \implies x=1-\frac{r}{3}.$$

$$(3)-(2):\ (y-4)^2-y^2=(r+3)^2-(r+1)^2 \implies -8y+16=4r+8 \implies y=1-\frac{r}{2}.$$

Now substitute x, y into equation (2):

$$x^2+y^2=(r+1)^2,$$

$$\left(1-\frac{r}{3}\right)^2+\left(1-\frac{r}{2}\right)^2=(r+1)^2.$$

Simplifying gives

$$23r^2+132r-36=0 \quad \implies \quad r=\boxed{\textbf{(B) } \frac{6}{23}}.$$

$\textbf{Remark.}$ This is a special case of $\text{Descartes' Kissing Theorem}$, which states that if 4 circles of curvature k_1, k_2, k_3, k_4 are mutually tangent to each other, then

$$(k_1+k_2+k_3+k_4)^2=2(k_1^2+k_2^2+k_3^2+k_4^2)$$